

RAPHAEL ANAADUMBA

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[GitHub](#) | [Google Scholar](#) | [Personal Website](#)

Computer science Ph.D. candidate studying when relational structure (neighboring homes, linked speech segments) improves prediction over tabular and sequence models, and when it does not. Five first-author papers (AAAI, ICASSP, Scientific Reports); graph models outperform XGBoost and spatial-only baselines for lead-risk prediction. Built pipelines from web scraping to a deployed LLM application; former backend engineer; NeurIPS 2026 reviewer.

RESEARCH EXPERIENCE

University of Massachusetts Lowell | CUBICS Lab

2023 - Present

Graduate Research Assistant | Lowell, MA

- Led development of graph neural network models for lead-contamination risk prediction on a \$2.5M NSF-funded project, working with 50,000+ water-test records across Chicago, Flint, and Massachusetts.
- Integrated Census, American Community Survey, and EPA data into spatial graphs; built a scraper and address-matching pipeline for masked Massachusetts EEA records (635 facilities, 99.5% address coverage).
- Diagnosed shortcut learning in a Massachusetts lead-risk model: a pooled AUROC of 0.78 concealed within-city AUROC of 0.34–0.62 because features encoded city identity, including a Simpson’s paradox in assessed value. Moved evaluation to per-city metrics and fixed a cross-validation split that placed all single-sample cities in one fold.
- Developing a graph-conditioned latent diffusion model (VAE, GIN encoder, DDPM) that generates synthetic training records, to measure how much real labeled data it can replace across cities.
- Developed reproducible workflows for data ingestion, record linkage, graph construction, ablations, threshold selection, and GIS risk maps; presented findings to the U.S. EPA and Merrimack River Watershed Council.

SELECTED PUBLISHED RESEARCH

Neighborhood-scale lead-risk prediction with spatial GATs

AAAI | IAAI Track

2026

- Aggregated 38,000+ EPA water tests into 739 census block groups and learned spatial edge weights; 91.6% recall for unsafe locations (0.83 AUROC) with 44% fewer false positives than spatial-only baselines.

Dementia detection from speech with AST and GAT

ICASSP

2026

- Combined Audio Spectrogram Transformer features with speaker–utterance graphs, achieving 80.5% accuracy on the Pitt Corpus and 81.8% on TAUADIAL; ablations isolate temporal connections as the main signal for MCI detection.

Community-engaged lead-risk assessment with self-supervised GATs

Scientific Reports

2026

- Achieved 90% accuracy and 0.836 AUROC by combining self-supervised pretraining with graph attention networks using public housing records and historical water-testing data.

Respiratory disease classification with graph convolutional networks

IEEE/ACM CHASE

2025

- Represented lung-sound recordings as relational graphs and developed a GCN using synthesized non-speech audio features; evaluated across respiratory conditions using 1,000+ clinical recordings.

Household lead-risk prediction in Flint with graph neural networks

Frontiers in Environmental Engineering

2025

- Linked parcel, Zillow, and water-test records into a dataset of 916 Flint homes (from ~14,000 tests); the GAT reached 0.93 AUROC and 0.93 recall versus 0.67 recall for XGBoost, using housing characteristics.

PROJECTS

Research Graph Explorer | [Live demo](#)

Deployed LLM application | JavaScript, D3.js, Vercel

- Built a retrieval-augmented question-answering tool over my publications: semantic-embedding retrieval, a D3.js knowledge graph, and a Groq-hosted LLM behind a Vercel edge function that keeps the API key server-side.

INDUSTRY EXPERIENCE

Shenzhen Roynard Service Consulting Co.

2021 - 2022

Software Engineer | Backend Engineering | Shenzhen, China

- Implemented Node.js backend services and REST APIs for authentication, role management, and business-process automation for enterprise clients.
- Refactored asynchronous data-ingestion routes and database indexes to reduce request latency under concurrent load.

TEACHING EXPERIENCE

University of Massachusetts Lowell

Fall 2026

Graduate Teaching Assistant | COMP 3050 Computer Architecture | Lowell, MA

- Set up the course's assignment-submission system; grade seven programming assignments for 40+ undergraduates.

EDUCATION

Ph.D. in Computer Science | Lowell, MA

University of Massachusetts Lowell

Expected May 2027

Research: Graph neural networks for environmental health and clinical audio.

Advisor: Dr. Mohammad Arif Ul Alam | NSF Award #2230180

M.S. in Computer Science (Machine Learning) | Nanjing, China

Nanjing University of Information Science & Technology

2020

B.S. in Computer Science | Tamale, Ghana

University for Development Studies

2016

TECHNICAL SKILLS

Programming and systems: Python, SQL, C++, Java, MATLAB, JavaScript, Node.js, Linux/Bash, Git, Docker

Machine learning: PyTorch, PyTorch Geometric, TensorFlow, scikit-learn, XGBoost, TabPFN, Hugging Face

Methods: GNNs (GAT, GCN, GraphSAGE, GIN), self-supervised learning, diffusion models, multimodal fusion, RAG

Data and geospatial: pandas, NumPy, NetworkX, GeoPandas, geopy, Selenium, QGIS, ArcGIS, REST APIs

AWARDS AND ACADEMIC SERVICE

Conference reviewer: NeurIPS, Evaluations & Datasets Track

2026

Journal reviewer: Scientific Reports, The Journal of Supercomputing, Discover Sustainability; completed 4+ reviews.

Competition: Top-three finalist, SAND Challenge: dysarthria severity classification from speech (AST + GAT).

Travel award: NSF Student Travel Award, IEEE/ACM CHASE, New York, NY

2025